Answer Key - Engineering - Graduate Level Assessment 7 (Answers are ordered exactly as in the question paper.)

# Multiple Choice

1. **Answer:** D
   *Options:* A) f(t) = 1 + 2 $^{\infty}\sum\_k = 1[\{coskt\}/\{1{+}k^2\}]$; B) f(t) = $^{\infty}\sum\_k = 1[\{coskt + ksinkt\}/\{1{+}k^2\}]$; C) f(t) = 2 $^{\infty}\sum\_k = 1[\{coskt + ksinkt\}/\{1{+}k^2\}]$; D) f(t) = 1 + 2 $^{\infty}\sum\_k = 1[\{coskt + ksinkt\}/\{1{+}k^2\}]$; E) f(t) = 2 + 2 $^{\infty}\sum\_k = 1[\{coskt\}/\{1{+}k^2\}]$
   *Note:* Correct option: D - f(t) = 1 + 2 $^{\infty}\sum\_k = 1[\{coskt + ksinkt\}/\{1{+}k^2\}]$

2. **Answer:** C
   *Options:* A) U(t) = L_0 (1 - tanht)cos + _0 (t - $sech^2$ t)sin; B) U(t) = L_0 (1 + $sech^2$ t)cos+ _0 (t +tanht)sin; C) U(t) = L_0 (1 + $sech^2$ t)cos- _0 (t +tanht)sin; D) U(t) = L_0 (1 - $sech^2$ t)cos- _0 (t +tanht)sin; E) U(t) = -L_0 (1 + $sech^2$ t)sin + _0 (t + tanht)cos
   *Note:* Correct option: C - U(t) = L_0 (1 + $sech^2$ t)cos- _0 (t +tanht)sin

3. **Answer:** D
   *Options:* A) 15 cycles/sec; B) 33 cycles/sec; C) 22.5 cycles/sec; D) 25.8 cycles/sec; E) 18 cycles/sec
   *Note:* Correct option: D - 25.8 cycles/sec

4. **Answer:** B
   *Options:* A) 28,500 Btu; B) 34,650 Btu; C) 40,000 Btu; D) 25,000 Btu; E) 45,000 Btu
   *Note:* Correct option: B - 34,650 Btu

5. **Answer:** A
   *Options:* A) Multiplexers; B) Memory Cache; C) Registers; D) Flip-Flops; E) Buffers
   *Note:* Correct option: A - Multiplexers

# True / False

6. **Answer:** False
   *Options:* A) True; B) False
   *Note:* LLM-generated false statement. Correct answer: 'high and low'.

7. **Answer:** False
   *Options:* A) True; B) False
   *Note:* LLM-generated false statement. Correct answer: '[k / ($s^2$ + $k^2$)]'.

8. **Answer:** True
   *Options:* A) True; B) False
   *Note:* LLM-generated true statement based on MCQ.

9. **Answer:** False
   *Options:* A) True; B) False
   *Note:* LLM-generated false statement. Correct answer: 'First digit from left to right'.

10. **Answer:** False
    *Options:* A) True; B) False
    *Note:* LLM-generated false statement. Correct answer: 'Hartley oscillator'.

# Long Form Response

11. **Answer:** $2.184 \times 10^-10 kg/sec.Explainwhythisisthecorrectanswerandprovidesupportingevidence.$
    $Note: Long-formderivedfromMCQ; awardcreditforstatingthecorrectideaandgivingsupportingrea$

11. **Answer:** (3/2) $(e^-t-e^-3t)V.Explainwhythisisthecorrectanswerandprovidesupportingevidence.Re$
    $Note: Long-formderivedfromMCQ; awardcreditforstatingthecorrectideaandgivingsupportingrea$

*End of Answer Key*